

◆ INNOVATION PROJECT PRESENTATION

Urban Glow-Grid

An IoT-Powered Interactive City Energy Installation

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The power grid is a **shared** responsibility.

Urban Glow-Grid goes beyond the simple message of "save electricity." It makes invisible energy data **tangible and immediate** — showing citizens that every individual choice ripples across the entire city, affecting hospitals, schools, and neighbours alike.

CIVIC AWARENESS

Users experience first-hand how collective decisions shape urban resilience and power stability.

REAL-TIME CONSEQUENCES

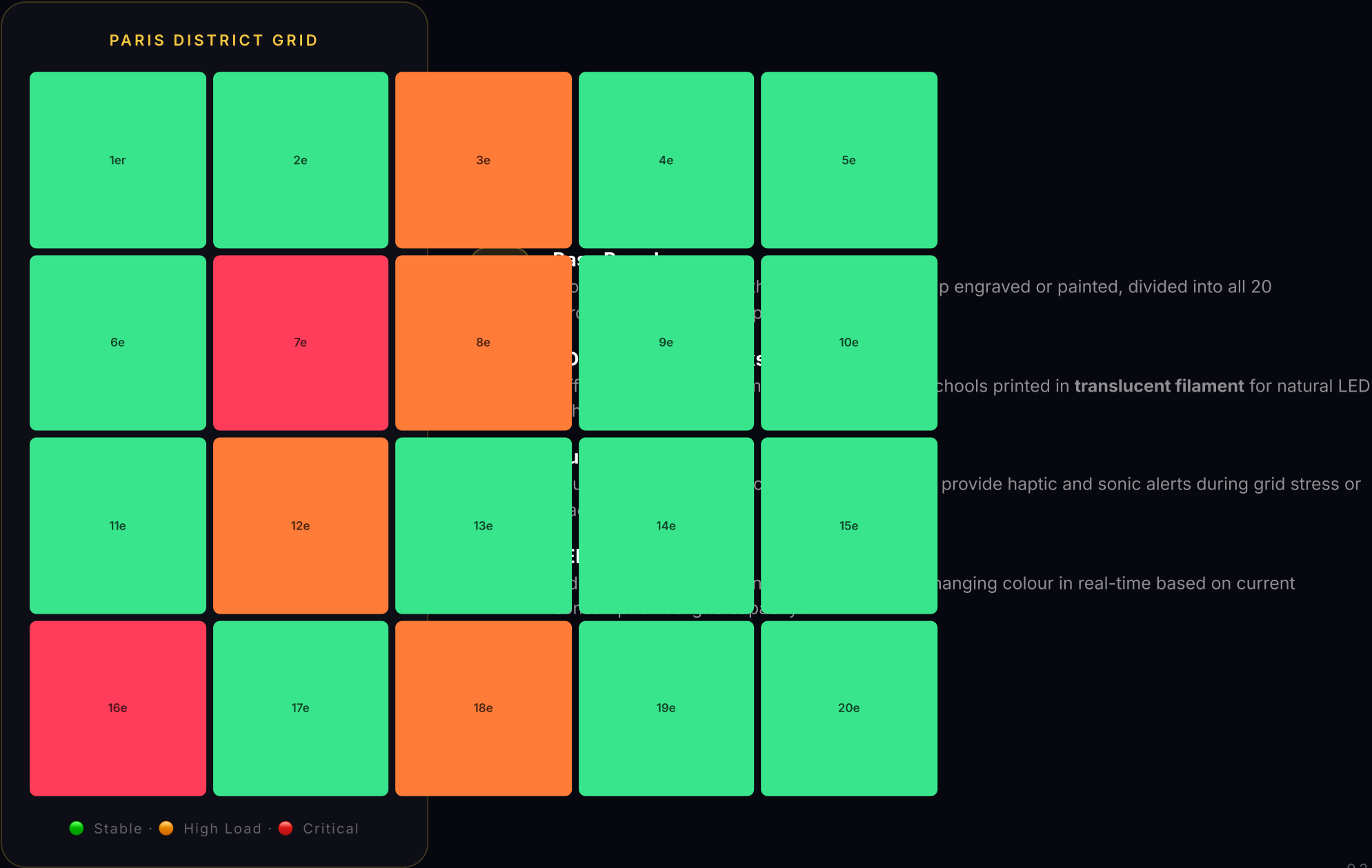
Decisions are immediately reflected in LED states and audio feedback — no abstraction, pure consequence.

EDUCATIONAL IMPACT

Designed to be exhibited at schools, fairs, and innovation events to spark lasting behavioural change.

Physical Design — La Maquette

A low-cost ($\leq 25\text{€}$) functional model of Paris with 20 independent zones



Hardware Architecture & IoT Mechanisms



The Brain — ESP32 / Arduino

The central microcontroller manages all district states, reads sensor inputs, controls the LED matrix, and triggers buzzer alerts in real-time.



Direct Interaction — Physical Buttons

Buttons mounted directly on the maquette allow interaction with priority buildings — toggle a school's lights to redirect power to a hospital.



Control Panel — Toggle Switches

A dedicated operator interface with labelled toggle switches lets users activate or deactivate individual districts, simulating real grid management.



IoT Connectivity

Optional Wi-Fi/BLE module on the ESP32 enables future remote monitoring, scenario upload, and data logging via a connected dashboard.

GRID STATUS INDICATORS



Green — Balanced

All zones powered.
Consumption within safe threshold.



Orange — High Load

Approaching limit. Rebalancing required soon.



Red / Flashing — Critical

Imminent blackout. Buzzers activate. Act now!

Gamification — The Experience

YOU ARE THE GRID MANAGER

1

THE CHALLENGE

Random Scenario Injection

The system triggers environmental scenarios — a cold winter night, a summer heatwave, or a major public event causing a demand surge. Each scenario has unique consumption profiles per district.

2

THE ACTION

Manual Load Distribution

Using toggles and buttons, the user redistributes power — cutting non-essential zones (commercial, decorative lighting) to maintain power for priority services like hospitals and emergency services.

3

THE CONSEQUENCE

Blackout or Balance

Exceed the safety threshold and the buzzers trigger a city-wide alarm. LEDs flash red. The city enters Blackout State — forcing a full restart and reflection on strategy.

EXAMPLE SCENARIOS



Cold Winter Night

All residential zones peak. Heating demand spikes across all arrondissements.



Summer Heatwave

Air conditioning overloads the southern districts. Risk of cascading failures.



National Event

Bastille Day surge — tourist zones and transport hubs consume maximum power.



BLACKOUT STATE

All LEDs cut. Buzzers sound. The experience resets — and the lesson is learned the hard way.

SDG Alignment & Social Impact

SDG 7

Affordable & Clean Energy

Urban Glow-Grid directly educates users on energy consumption patterns, grid stability constraints, and the role of renewable energy in preventing peak-hour overloads. By making these concepts visceral and interactive, it fosters long-term awareness of sustainable energy use.

- Visualises real-time grid load vs. capacity
- Demonstrates the value of demand-side management
- Encourages off-peak consumption habits
- Highlights the importance of green energy buffers

SDG 11

Sustainable Cities & Communities

The city-scale simulation models the interdependencies between urban zones — residential, commercial, healthcare, and education — revealing how smart urban planning and citizen participation are key to building resilient, livable, and energy-efficient cities.

- Models urban zone interdependencies
- Simulates priority-based resource allocation
- Promotes citizen participation in urban resilience
- Applicable in smart city planning education

Budget Breakdown & Project Timeline

COMPONENT BUDGET (≤ 25€)

ESP32 / Arduino microcontroller	~6€
LED strip / individual LEDs	~5€
Toggle switches & buttons	~4€
Piezo buzzers (×4)	~2€
MDF / wood board	~4€
3D printing filament (translucent)	~3€
Wiring, resistors, misc.	~1€
TOTAL ESTIMATED	~25€

MILESTONE TIMELINE

- **PHASE 1**
Concept & Design Validation
Finalize zone map, zoning logic, component list, and schematic diagrams
- **PHASE 2**
Hardware Prototyping
Assemble circuit, wire LEDs and buzzers, verify ESP32 firmware on breadboard
- **PHASE 3**
Physical Maquette Construction
3D print landmarks, mount on board, integrate LED network, build control panel
- **PHASE 4**
Game Logic & Scenario Programming
Implement challenge scenarios, threshold detection, blackout state, and buzzer triggers
- **PHASE 5**
Testing, Demo & Exhibition
Full integration test, UX review, and final showcase at the innovation event

◆ URBAN GLOW-GRID INNOVATION PROJECT

Meet the Team

Four students. One city. Infinite lessons about energy.



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IOT · GAMIFICATION · EDUCATION

INNOVATION SEMESTER · UTSEUS